

NEXUS Platform Hazard Assessment

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Platform:	NEXUS (All Products)
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1. Purpose

This document provides a platform-level hazard assessment for the NEXUS immunoassay system covering all products manufactured on shared production lines. It identifies hazards common to the NEXUS platform and maps them to product-specific FMEAs.

2. Hazard Identification

ID	Hazard Category	Hazard Description	Affected Products	Risk Level	Reference FMEA
H-001	Biological	Cross-contamination between antibody types	All NEXUS	HIGH	Product-specific FMEAs
H-002	Chemical	Reagent instability due to storage conditions	All NEXUS	MEDIUM	Product-specific FMEAs
H-003	Mechanical	sensor cartridge physical damage during assembly	All NEXUS	LOW	FMEA-CKMB-001, FMEA-B2M-001
H-004	Process	Assembly line cross-contamination during B2M, CK changeover	MB, Troponin	HIGH	FMEA-B2M-001
H-005	Process	Calibration signal drift from material variation	CK MB, Troponin	HIGH	FMEA-CKMB-001
H-006	Human	Operator error during manual assembly	All steps NEXUS	MEDIUM	Product-specific FMEAs
H-007	Environmental	Temperature excursion in storage areas	All NEXUS	MEDIUM	Product-specific FMEAs
H-008	IT/Software	ERP lot tracking data entry error	All NEXUS	LOW	Product-specific FMEAs

3. Risk Control Measures

HIGH risk hazards:

- H-001 (Cross-contamination): Dedicated sensor cartridges per product, barcode verification at assembly, SOP-ASM-003 changeover procedure.
- H-004 (Assembly line cross-contamination): Minimum 30-minute changeover, 8-item clearance checklist, visual verification of CART cartridge type.
- H-005 (Calibration signal drift): Incoming inspection per SOP-CART-001, antibody density measurement, trend monitoring of S1/S2 signals.

MEDIUM risk hazards:

- H-002 (Reagent instability): 2-8 C storage, continuous temperature monitoring.
- H-006 (Operator error): Qualification programme, annual retraining.
- H-007 (Temperature excursion): Alarmed storage, backup power systems.

4. Revision History

Version	Date	Author	Changes
1.0	2020-01-15	Quality Engineering	Initial release
2.0	2022-06-01	Quality Engineering	Added process hazards H-004, H-005 following INV-2022-0045
3.0	2023-06-01	Quality Engineering	Updated controls for H-001 and H-005; aligned with updated FMEAs

